

Introduction to Machine Learning

Course code-CSE475

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K-MEANS CLUSTERING

What is clustering?

Clustering is the classification of objects into different groups, or more precisely, the partitioning of a data set into subsets (clusters), so that the data in each subset (ideally) share some common trait - often according to some defined distance measure.

A Simple example showing the implementation of k-means algorithm (using K=2)

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Step 1: Randomly we choose following two centroids (k=2) for two clusters.
In this case the 2 centroid are: m1=(1.0,1.0) and m2=(5.0,7.0).

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Mean Vector	
Group 1	(1.0, 1.0)
Group 2	(5.0, 7.0)

	Mean Vector
Group 1	(1.0, 1.0)
Group 2	(5.0, 7.0)

Item	Variable 1	Variable 2	Centroid 1	Centroid 2
1	1	1	0	7.21
2	1.5	2	1.12	6.1
3	3	4	3.61	3.61
4	5	7	7.21	0
5	3.5	5	4.72	2.5
6	4.5	5	5.32	2.06
7	3.5	4.5	4.3	2.92

$$m_1 = \left(\frac{1}{3}(1.0 + 1.5 + 3.0), \frac{1}{3}(1.0 + 2.0 + 4.0) \right) = (1.83, 2.33)$$

$$m_2 = \left(\frac{1}{4}(5.0 + 3.5 + 4.5 + 3.5), \frac{1}{4}(7.0 + 5.0 + 5.0 + 4.5) \right) = (4.12, 5.38)$$

Step 2:

Thus, we obtain two clusters containing:

{1,2,3} and {4,5,6,7}.

• Their new centroids are:

	Mean Vector
Group 1	(1.83, 2.33)
Group 2	(4.12, 5.38)

Item	Variable 1	Variable 2	Centroid 1	Centroid 2
1	1	1	1.57	5.38
2	1.5	2	0.47	4.28
3	3	4	2.04	1.78
4	5	7	5.64	1.84
5	3.5	5	3.15	0.73
6	4.5	5	3.78	0.54
7	3.5	4.5	2.74	1.08

m1=(1.25,1.5)

New centroids are:

m2 = (3.9,5.1)

Therefore, the new clusters are:
 {1,2} and {3,4,5,6,7}

	Mean Vector
Group 1	$(1.25, 1.5)$
Group 2	$(3.9, 5.1)$

Item	Variable 1	Variable 2	Centroid 1	Centroid 2
1	1	1	0.56	5.02
2	1.5	2	0.56	3.92
3	3	4	3.05	1.42
4	5	7	6.66	2.2
5	3.5	5	4.16	0.41
6	4.5	5	4.78	0.61
7	3.5	4.5	3.75	0.72

New centroids are:

$m_1 = (1.25, 1.5)$

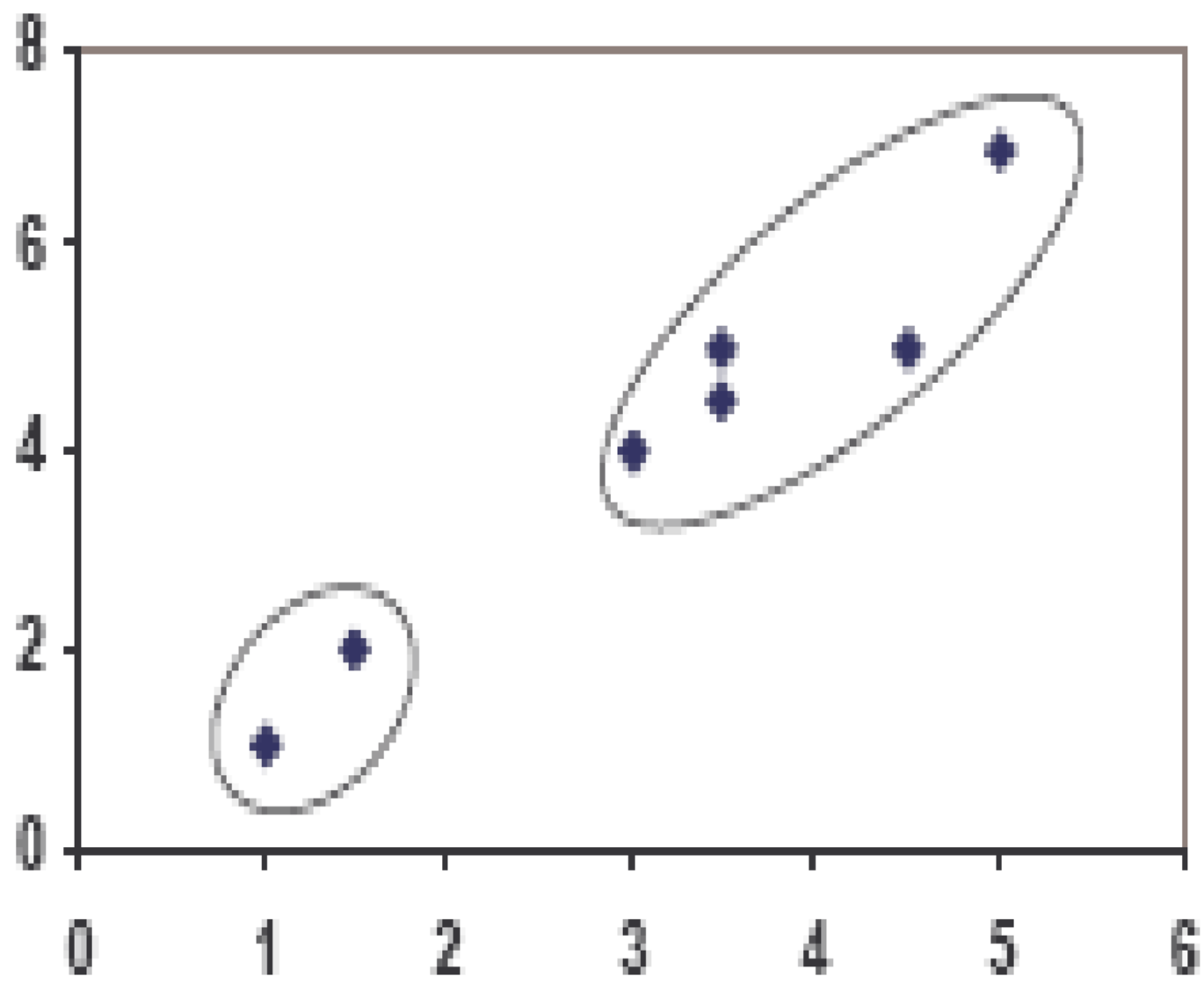
$m_2 = (3.9, 5.1)$

Therefore, the new clusters are:
 $\{1, 2\}$ and $\{3, 4, 5, 6, 7\}$

Step 4 :

- I. Therefore, there is no change in the cluster.
- II. Thus, the algorithm comes to a halt here and final result consist of 2 clusters $\{1,2\}$ and $\{3,4,5,6,7\}$.

Plot



What will happen when $K=3$?